

Rescaling Invariance and Its Failure in Econometrics: OLS Unit Invariance and Wald Test Non-Invariance Under Reparameterization

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Abstract—Continuing the series, we examine econometrics, the field where the invariance/non-invariance boundary has been studied most explicitly. We prove that ordinary least squares t -statistics are exactly invariant under linear rescaling of regressors and the dependent variable (a change of measurement units), and we prove and simulate the classical Dagenais–Dufour result that the Wald test statistic for a nonlinear restriction is *not* invariant to how that restriction is algebraically written, even when the restriction is mathematically equivalent. The latter produces a real, sizable numerical discrepancy and a measurable size distortion in finite samples, giving the sharpest illustration in this series of a genuine, provable non-invariance.



1 OLS COEFFICIENTS AND TEST STATISTICS: UNIT INVARIANCE

Consider the simple linear regression $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, estimated by OLS. Applied econometrics constantly changes units — dollars to thousands of dollars, people to millions of people — and a sound method must not have its statistical conclusions depend on that choice.

Proposition 1. *Let $x'_i = cx_i$, $y'_i = dy_i$ for constants $c, d > 0$. Then the OLS t -statistic on the slope coefficient satisfies $t'_{\beta_1} = t_{\beta_1}$ exactly, while $\hat{\beta}'_1 = (d/c)\hat{\beta}_1$ and $\widehat{\text{se}}(\hat{\beta}'_1) = (d/c)\widehat{\text{se}}(\hat{\beta}_1)$.*

Proof. OLS on rescaled data solves the same normal equations with $X' = cX$, $y' = dy$ (for the slope block), giving $\hat{\beta}'_1 = \frac{\text{Cov}(x', y')}{\text{Var}(x')} = \frac{cd \text{Cov}(x, y)}{c^2 \text{Var}(x)} = \frac{d}{c} \hat{\beta}_1$. Residuals scale as $\hat{\varepsilon}'_i = d\hat{\varepsilon}_i$ (since $y'_i - \hat{\beta}'_0 - \hat{\beta}'_1 x'_i = d(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$ term by term), so $\hat{\sigma}'^2 = d^2 \hat{\sigma}^2$, and $\widehat{\text{se}}(\hat{\beta}'_1) = \hat{\sigma}' / \sqrt{\sum (x'_i - \bar{x}')^2} = d\hat{\sigma} / (c\sqrt{\sum (x_i - \bar{x})^2}) = (d/c)\widehat{\text{se}}(\hat{\beta}_1)$. The t -statistic is therefore $t' = \hat{\beta}'_1 / \widehat{\text{se}}(\hat{\beta}'_1) = \frac{(d/c)\hat{\beta}_1}{(d/c)\widehat{\text{se}}(\hat{\beta}_1)} = t_{\beta_1}$, since d/c cancels exactly. $\square$

1.1 Worked numerical verification

Simulating $n = 2,000$ observations from $y = 3.0 + 1.7x + \varepsilon$ gives $\hat{\beta}_1 = 1.69544826$, $\text{se} = 0.01127119$, $t = 150.42317144$. Rescaling x by $c = 1,000$ (e.g., dollars to thousandths) and y by $d = 0.001$ gives $\hat{\beta}'_1 = 1.69544826 \times 10^{-6}$ and $\text{se}' = 1.12711907 \times 10^{-8}$ — both individually far smaller, exactly as predicted — but the t -statistic is 150.42317144, matching the original to all displayed digits. This is the same structural fact proved for linear rescaling throughout this series, now confirmed for regression inference specifically: a change of measurement units changes coefficient magnitudes but never the statistical significance conclusion.

2 WHERE INVARIANCE FAILS: THE WALD TEST UNDER REPARAMETERIZATION

Not every quantity built from estimated parameters is invariant to how those parameters are written. The clearest and most consequential counterexample in econometrics is the Wald test applied to a nonlinear restriction.

Consider testing $H_0 : \beta = 1$ in the model $y_i = \beta x_i + \varepsilon_i$. This null is mathematically equivalent to $H_0 : \gamma = 1$ where $\gamma \equiv 1/\beta$ (for $\beta \neq 0$) — the same restriction, restated. A Wald test can be built either way.

Proposition 2. *Let $\hat{\beta}$ be the OLS estimator with asymptotic variance V_β , and let $\hat{\gamma} = 1/\hat{\beta}$ with delta-method variance $V_\gamma = (1/\hat{\beta}^2)^2 V_\beta$. The two Wald statistics*

$$W_\beta = \frac{(\hat{\beta} - 1)^2}{V_\beta}, \quad W_\gamma = \frac{(\hat{\gamma} - 1)^2}{V_\gamma} \quad (1)$$

test the identical null hypothesis but are not equal for finite samples, and only coincide asymptotically as $n \rightarrow \infty$.

Proof sketch. Substituting $\hat{\gamma} = 1/\hat{\beta}$ and $V_\gamma = V_\beta/\hat{\beta}^4$:

$$W_\gamma = \frac{(1/\hat{\beta} - 1)^2}{V_\beta/\hat{\beta}^4} = \frac{\hat{\beta}^4(1 - \hat{\beta})^2/\hat{\beta}^2}{V_\beta} = \frac{\hat{\beta}^2(1 - \hat{\beta})^2}{V_\beta} = \hat{\beta}^2 W_\beta. \quad (2)$$

$W_\gamma = W_\beta$ exactly only when $\hat{\beta}^2 = 1$; in general $W_\gamma \neq W_\beta$, with the discrepancy shrinking only as $\hat{\beta} \rightarrow 1$ under H_0 in large samples (where the delta method becomes exact). This is the Dagenais–Dufour non-invariance result: the Wald statistic depends on an arbitrary modeling choice — how the restriction is algebraically parameterized — not just on the restriction’s content. $\square$

2.1 Worked numerical verification

With true slope $\beta = 1.6$, $n = 60$, testing the false null $H_0 : \beta = 1$: a single sample gives $\hat{\beta} = 1.7366$, yielding

$W_\beta = 28.07$ under the direct parameterization but $W_\gamma = 84.66$ under the reciprocal parameterization — a factor of $3.0\times$ difference for testing the identical hypothesis on the identical data.

Averaging over 5,000 independent samples confirms this is systematic, not a fluke of one draw: mean $W_\beta = 23.19$ versus mean $W_\gamma = 63.52$, and in 100% of the 5,000 samples the two statistics differed by more than 10%.

The practical consequence — test size distortion — is smaller but still measurable. Simulating under the *true* null $\beta = 1$ (so a correctly-sized 5% test should reject 5% of the time), the β -parameterized Wald test rejects at an empirical rate of 4.92% (correctly calibrated), while the γ -parameterized version rejects at 5.56% — a real, if modest, inflation of the false-positive rate purely from an algebraically equivalent restatement of the same null hypothesis.

TABLE 1
Wald test non-invariance, simulated ($n = 60$)

	β -parameterization	$\gamma = 1/\beta$ -parameterization
Mean W (false null, $\beta_{\text{true}} = 1.6$)	23.19	63.52
Empirical rejection rate (true null, nominal 5%)	4.92%	5.56%

3 CONCLUSION

Econometrics gives the sharpest confirmation in this series of both sides of the same underlying question:

- OLS inference (Section I) is provably, exactly invariant to the choice of measurement units — confirmed to all displayed digits even under a combined 10^6 -factor rescaling of x and y .
- The Wald test under a nonlinear reparameterization (Section II) is *not* invariant — confirmed by simulation to differ by a factor of $\sim 3\times$ on typical finite samples, and to distort test size measurably even under the true null.

As throughout this series, the two results are not in tension: linear rescaling of the data is a bijection preserving all relative structure (Section I, and the entire earlier arc of this series); reparameterizing a restriction through a *nonlinear* function of the estimator is a fundamentally different operation, and the Wald statistic — unlike the likelihood ratio or score statistics, which remain invariant to reparameterization — was never guaranteed to survive it. Knowing which is which, precisely, is the entire content of this series.

REFERENCES

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